

R Program codes

1.Smart City Air Quality Monitoring

➤ Sequential Color Palette

```
library(ggplot2)
```

```
air <- data.frame(
```

```
  Location = 1:5,
```

```
  PM25 = c(30, 60, 90, 50, 80)
```

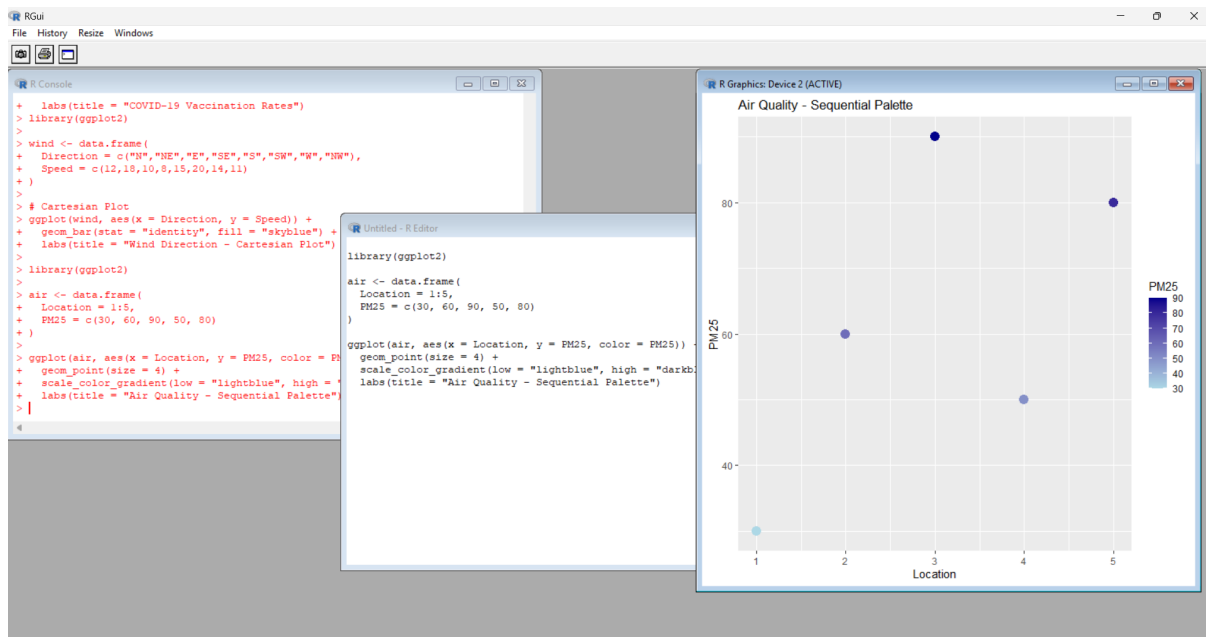
```
)
```

```
ggplot(air, aes(x = Location, y = PM25, color = PM25)) +
```

```
  geom_point(size = 4) +
```

```
  scale_color_gradient(low = "lightblue", high = "darkblue") +
```

```
  labs(title = "Air Quality - Sequential Palette")
```



➤ Diverging Color Palette

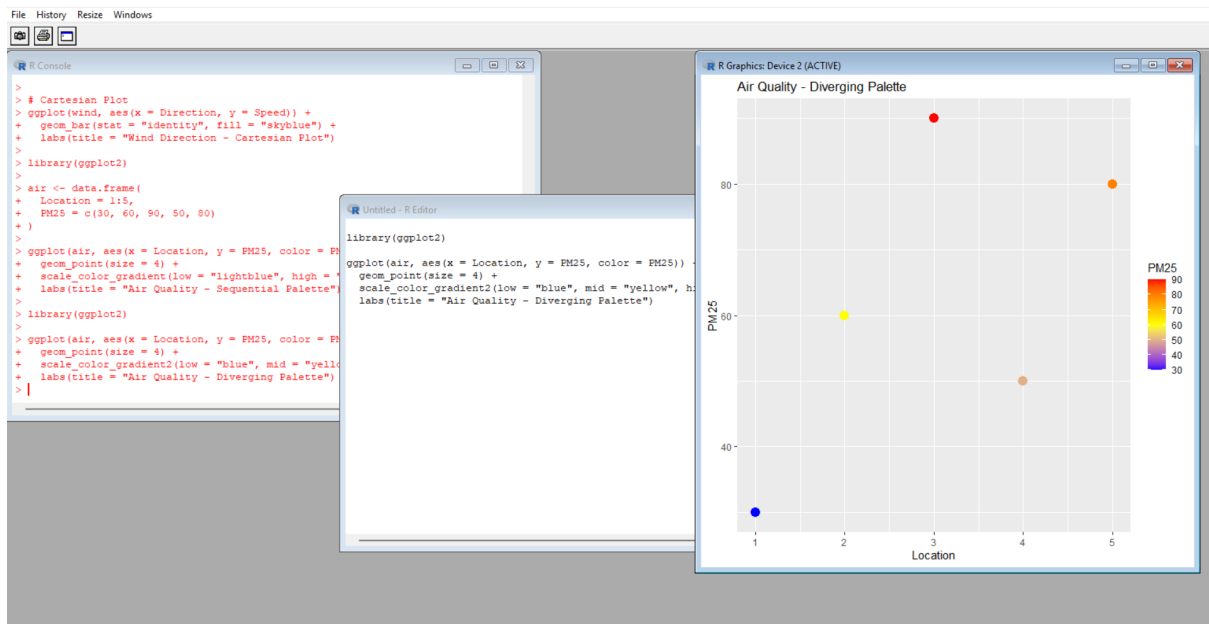
```
library(ggplot2)
```

```
ggplot(air, aes(x = Location, y = PM25, color = PM25)) +
```

```
  geom_point(size = 4) +
```

```
  scale_color_gradient2(low = "blue", mid = "yellow", high = "red", midpoint = 60) +
```

```
  labs(title = "Air Quality - Diverging Palette")
```



➤ Qualitative Color Palette

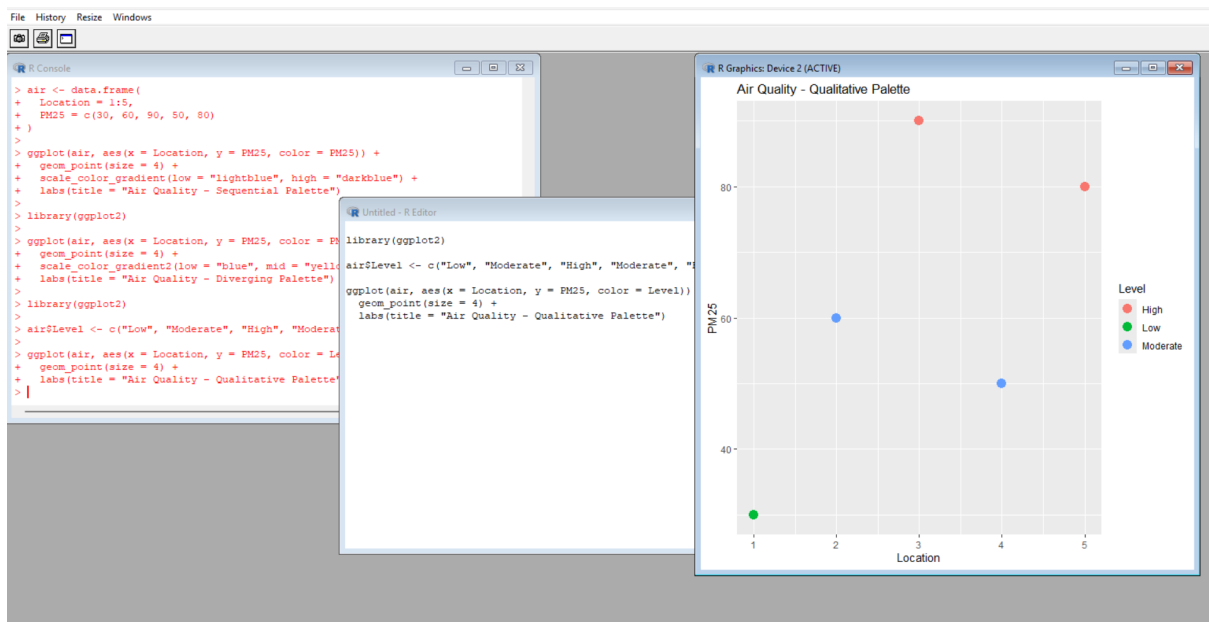
library(ggplot2)

```
air$Level <- c("Low", "Moderate", "High", "Moderate", "High")
```

```
ggplot(air, aes(x = Location, y = PM25, color = Level)) +
```

```
  geom_point(size = 4) +
```

```
  labs(title = "Air Quality - Qualitative Palette")
```

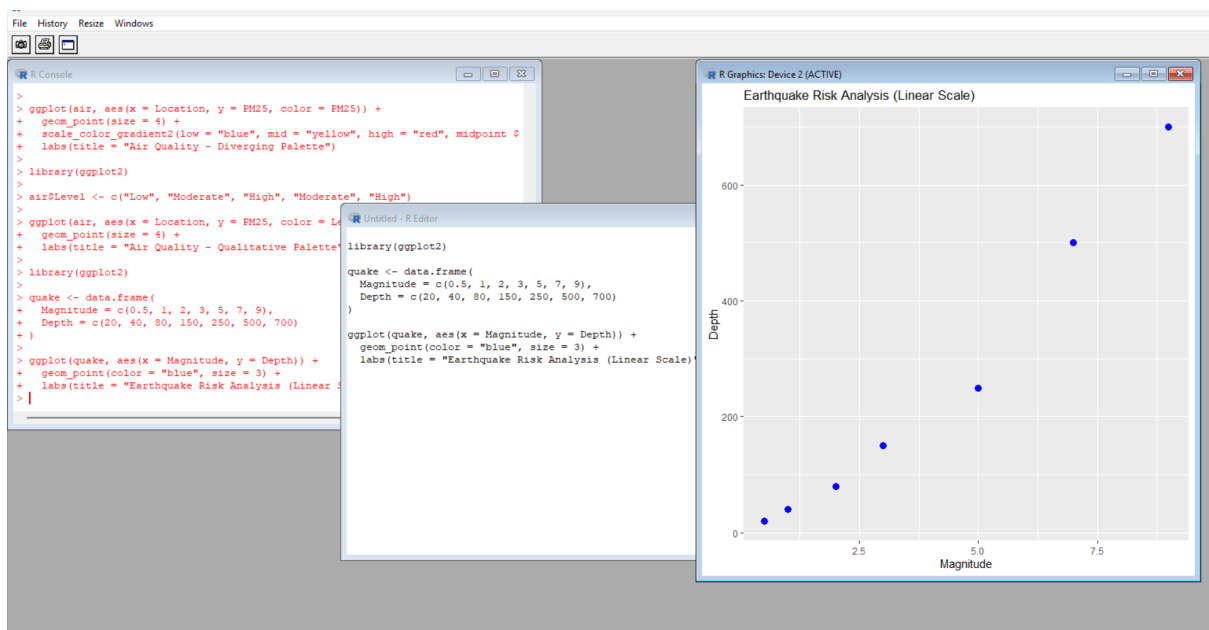


2. Earthquake Risk Analysis

➤ Linear Axis

```
library(ggplot2)
```

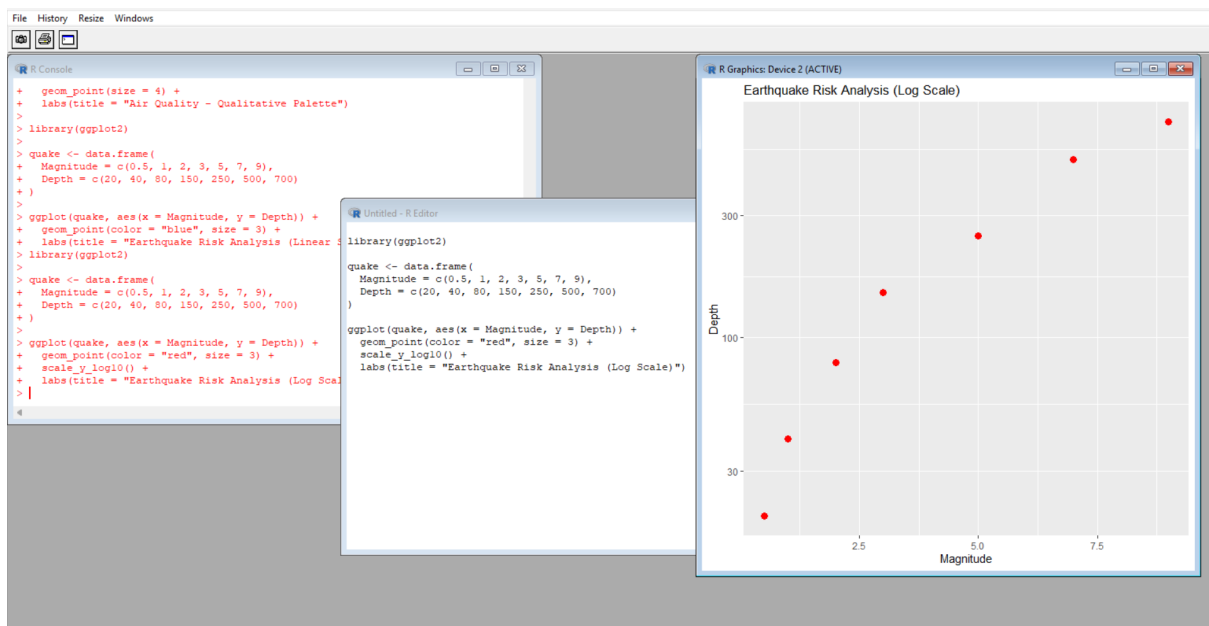
```
quake <- data.frame(  
  Magnitude = c(0.5, 1, 2, 3, 5, 7, 9),  
  Depth = c(20, 40, 80, 150, 250, 500, 700)  
)  
  
ggplot(quake, aes(x = Magnitude, y = Depth)) +  
  geom_point(color = "blue", size = 3) +  
  labs(title = "Earthquake Risk Analysis (Linear Scale)")
```



➤ Logarithmic Axis

```
library(ggplot2)
```

```
quake <- data.frame(  
  Magnitude = c(0.5, 1, 2, 3, 5, 7, 9),  
  Depth = c(20, 40, 80, 150, 250, 500, 700)  
)  
  
ggplot(quake, aes(x = Magnitude, y = Depth)) +  
  geom_point(color = "red", size = 3) +  
  scale_y_log10() +  
  labs(title = "Earthquake Risk Analysis (Log Scale)")
```

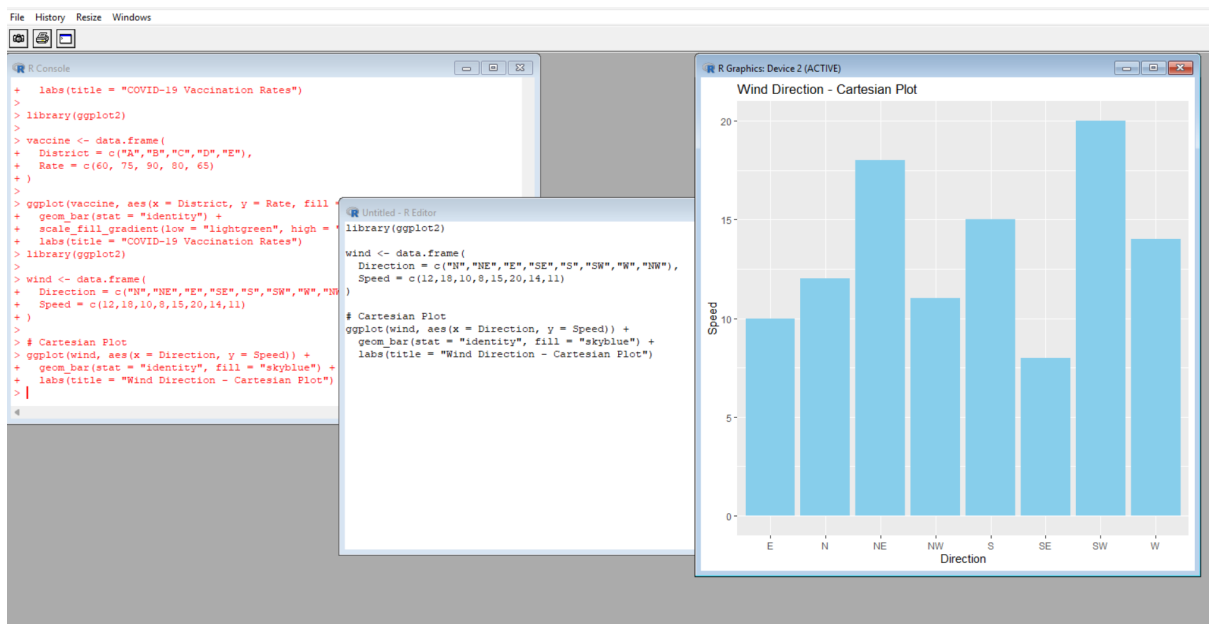


3. Wind Direction Analysis for Airport Operations

```
library(ggplot2)
```

```
wind <- data.frame(  
  Direction = c("N","NE","E","SE","S","SW","W","NW"),  
  Speed = c(12,18,10,8,15,20,14,11)  
)
```

```
ggplot(wind, aes(x = Direction, y = Speed)) +  
  geom_bar(stat = "identity", fill = "skyblue") +  
  labs(title = "Wind Direction - Cartesian Plot")
```



➤ Polar Coordinate (Wind Rose)

```
library(ggplot2)
```

```
wind <- data.frame(
```

```
  Direction = c("N","NE","E","SE","S","SW","W","NW"),
```

```
  Speed = c(12,18,10,8,15,20,14,11)
```

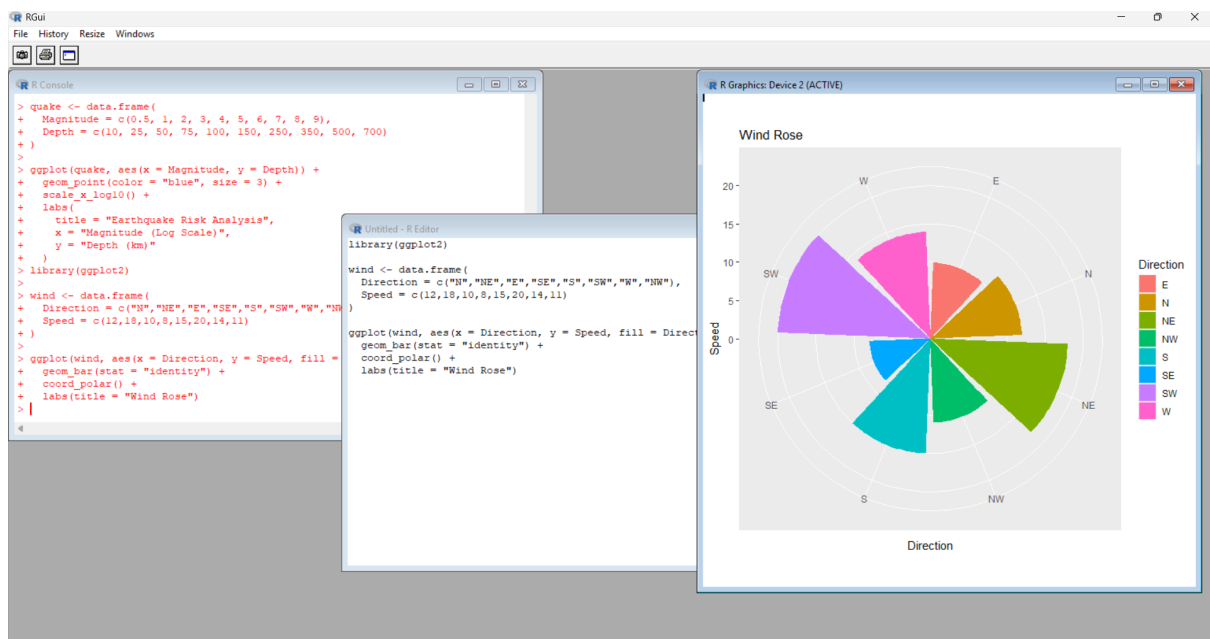
```
)
```

```
ggplot(wind, aes(x = Direction, y = Speed, fill = Direction)) +
```

```
  geom_bar(stat = "identity") +
```

```
  coord_polar() +
```

```
  labs(title = "Wind Rose")
```

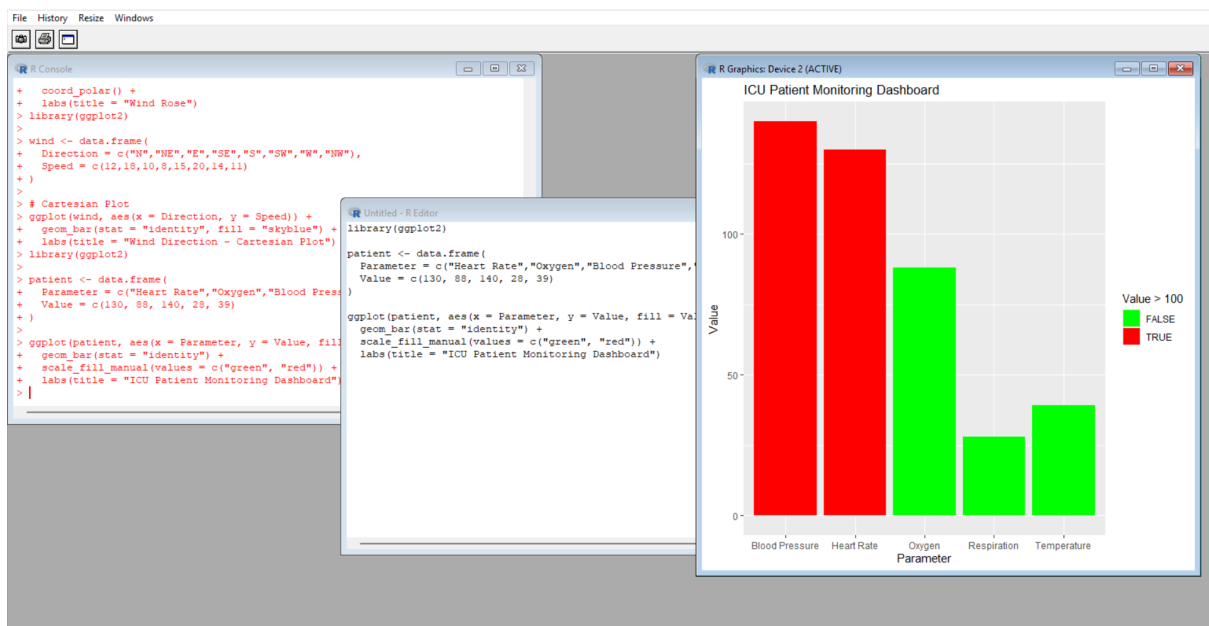


4. Hospital Patient Monitoring Dashboard

```
library(ggplot2)
```

```
patient <- data.frame(  
  Parameter = c("Heart Rate", "Oxygen", "Blood Pressure", "Respiration", "Temperature"),  
  Value = c(130, 88, 140, 28, 39)  
)
```

```
ggplot(patient, aes(x = Parameter, y = Value, fill = Value > 100)) +  
  geom_bar(stat = "identity") +  
  scale_fill_manual(values = c("green", "red")) +  
  labs(title = "ICU Patient Monitoring Dashboard")
```



5. Global Climate Change Study

```
library(ggplot2)
```

```
climate <- data.frame(
```

```
  Country = c("India","USA","China","Japan","UK"),
```

```
  Year = c(2021,2022,2023,2024,2025),
```

```
  Anomaly = c(-1.5,-0.5,0.0,1.2,2.0)
```

```
)
```

```
ggplot(climate, aes(x = Year, y = Country, fill = Anomaly)) +
```

```
  geom_tile() +
```

```
  scale_fill_gradient2(
```

```
    low = "blue",
```

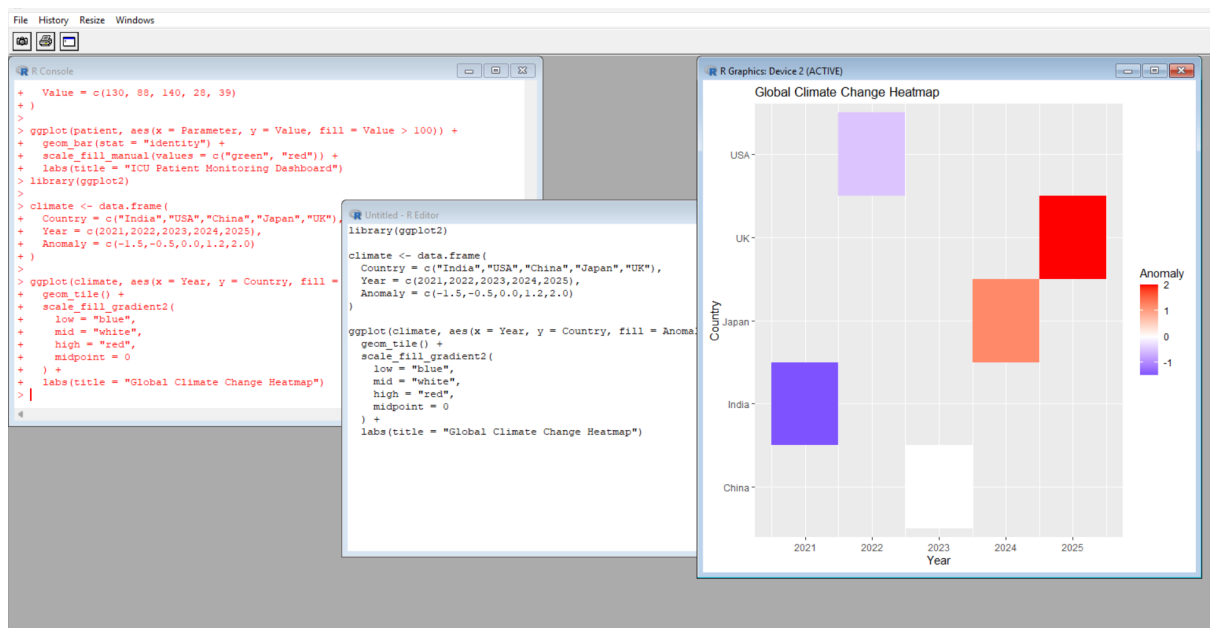
```
    mid = "white",
```

```
    high = "red",
```

```
    midpoint = 0
```

```
) +
```

```
labs(title = "Global Climate Change Heatmap")
```

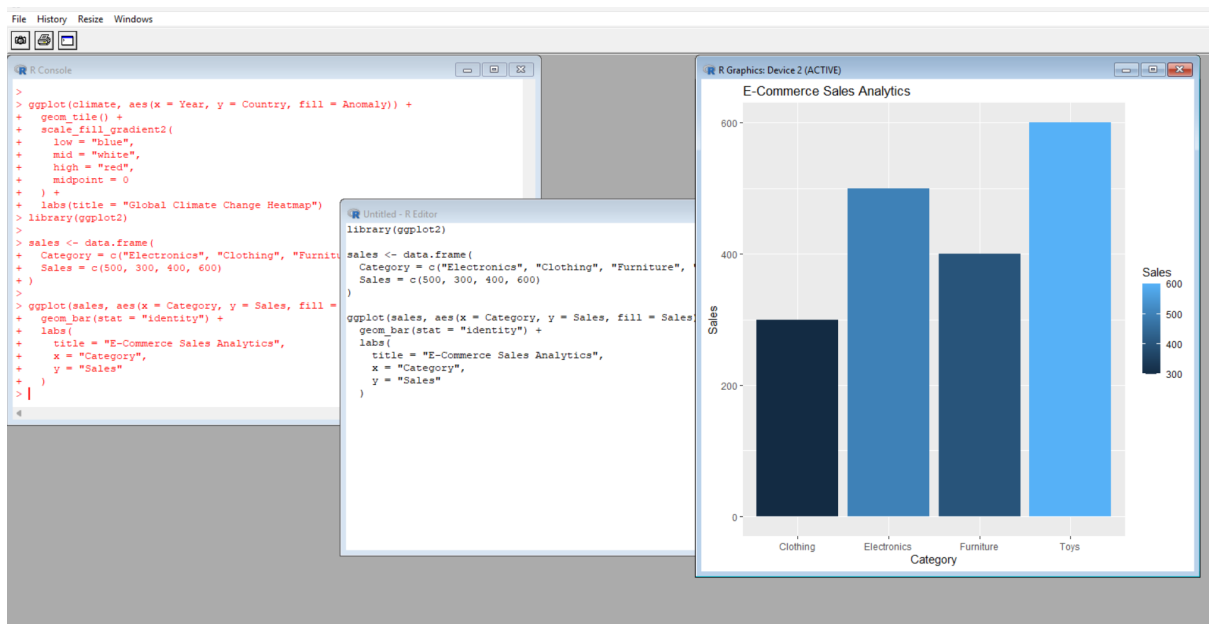


6. E-Commerce Sales Analytics

```
library(ggplot2)
```

```
sales <- data.frame(  
  Category = c("Electronics", "Clothing", "Furniture", "Toys"),  
  Sales = c(500, 300, 400, 600)  
)
```

```
ggplot(sales, aes(x = Category, y = Sales, fill = Sales)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "E-Commerce Sales Analytics",  
    x = "Category",  
    y = "Sales"  
  )
```



7. Satellite Image Vegetation Analysis

```
library(ggplot2)
```

```
ndvi <- data.frame(
```

```
  Area = c("A", "B", "C", "D", "E"),
```

```
  NDVI = c(-1, -0.5, 0, 0.5, 1)
```

```
)
```

```
ggplot(ndvi, aes(x = Area, y = NDVI, fill = NDVI)) +
```

```
  geom_bar(stat = "identity") +
```

```
  scale_fill_gradient2(low = "blue", mid = "yellow", high = "green", midpoint = 0) +
```

```
  labs(title = "Vegetation Analysis")
```



8. Cryptocurrency Market Volatility

```
library(ggplot2)
```

```
crypto <- data.frame(  
  Day = 1:5,  
  Bitcoin = c(30000, 35000, 40000, 45000, 50000),  
  Ethereum = c(2000, 2200, 2500, 2800, 3000)  
)  
  
ggplot(crypto, aes(x = Day)) +  
  geom_line(aes(y = Bitcoin, color = "Bitcoin")) +  
  geom_line(aes(y = Ethereum, color = "Ethereum")) +  
  scale_y_log10() +  
  labs(  
    title = "Cryptocurrency Market Volatility",  
    x = "Day",  
    y = "Price (Log Scale)"  
  )
```

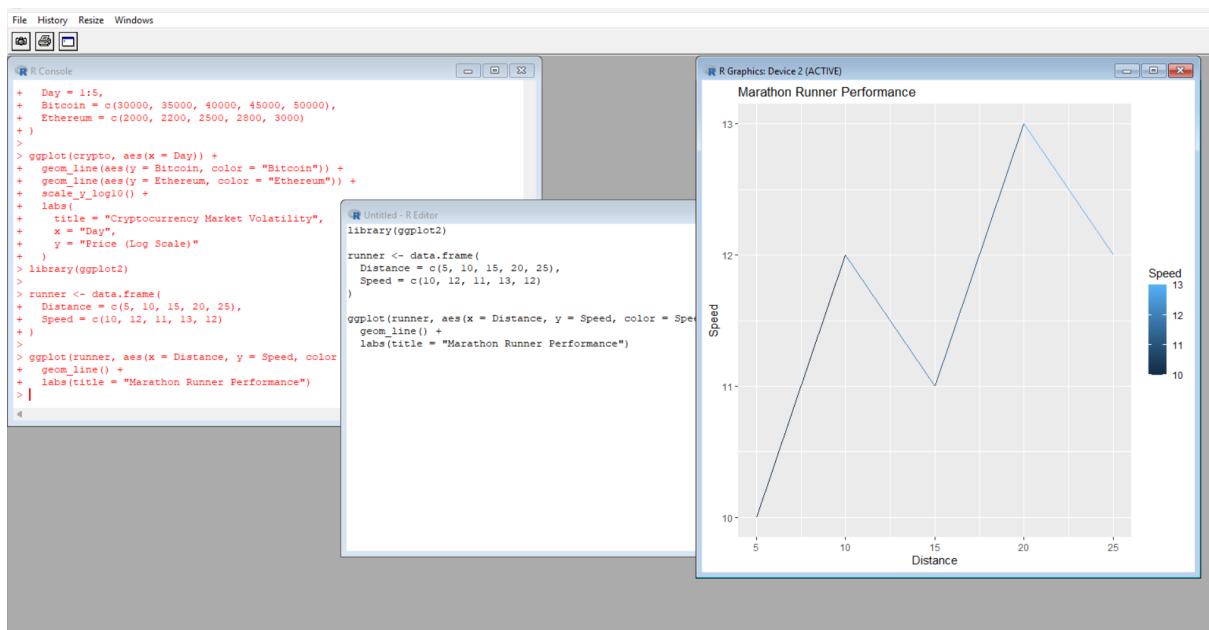


9. Marathon Runner Performance Analysis

➤ Cartesian Plot

```
library(ggplot2)
```

```
runner <- data.frame(  
  Distance = c(5, 10, 15, 20, 25),  
  Speed = c(10, 12, 11, 13, 12)  
)  
  
ggplot(runner, aes(x = Distance, y = Speed, color = Speed)) +  
  geom_line() +  
  labs(title = "Marathon Runner Performance")
```

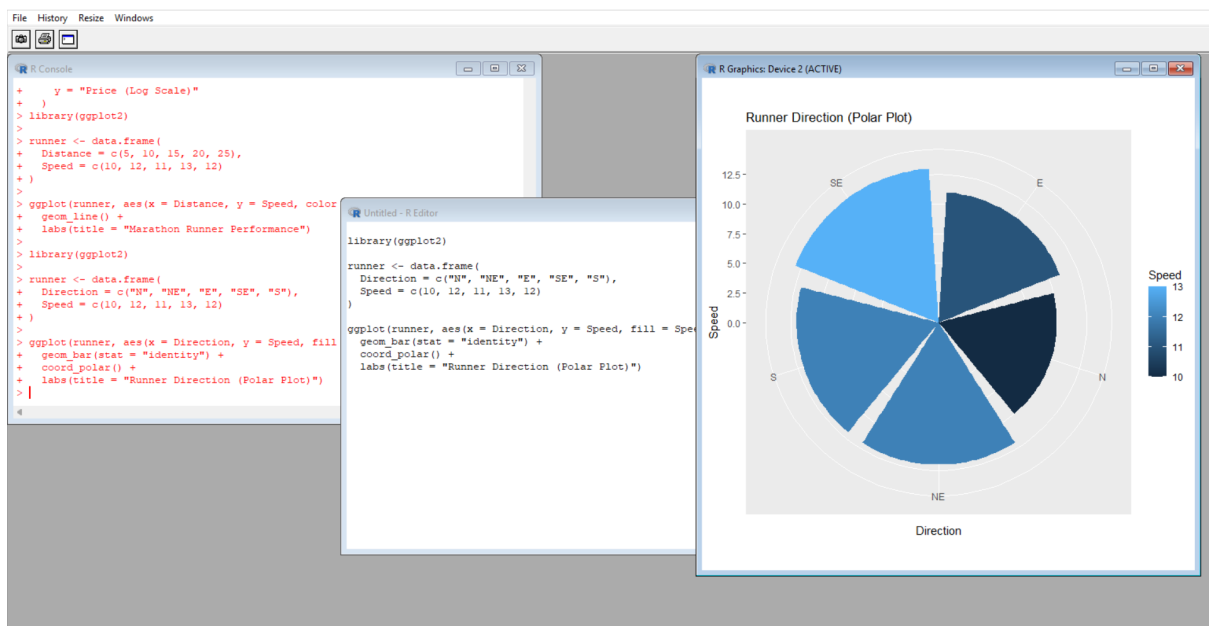


➤ Polar Coordinate Plot

```
library(ggplot2)
```

```
runner <- data.frame(  
  Direction = c("N", "NE", "E", "SE", "S"),  
  Speed = c(10, 12, 11, 13, 12)  
)
```

```
ggplot(runner, aes(x = Direction, y = Speed, fill = Speed)) +  
  geom_bar(stat = "identity") +  
  coord_polar() +  
  labs(title = "Runner Direction (Polar Plot)")
```



10. COVID-19 Vaccination Campaign Dashboard

Version 1: Policymakers Dashboard (Blue Palette)

```
library(ggplot2)
```

```
vaccine <- data.frame(
```

```
  District = c("A","B","C","D","E"),
```

```
  Rate = c(60, 75, 90, 80, 65)
```

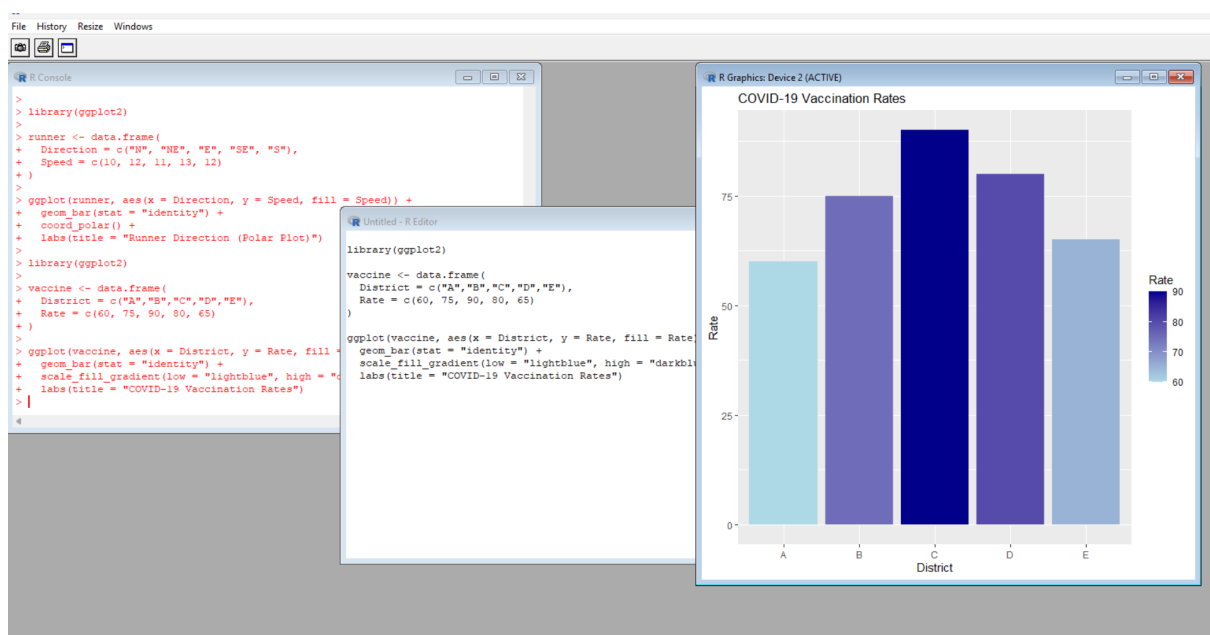
```
)
```

```
ggplot(vaccine, aes(x = District, y = Rate, fill = Rate)) +
```

```
  geom_bar(stat = "identity") +
```

```
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
```

```
  labs(title = "COVID-19 Vaccination Rates")
```

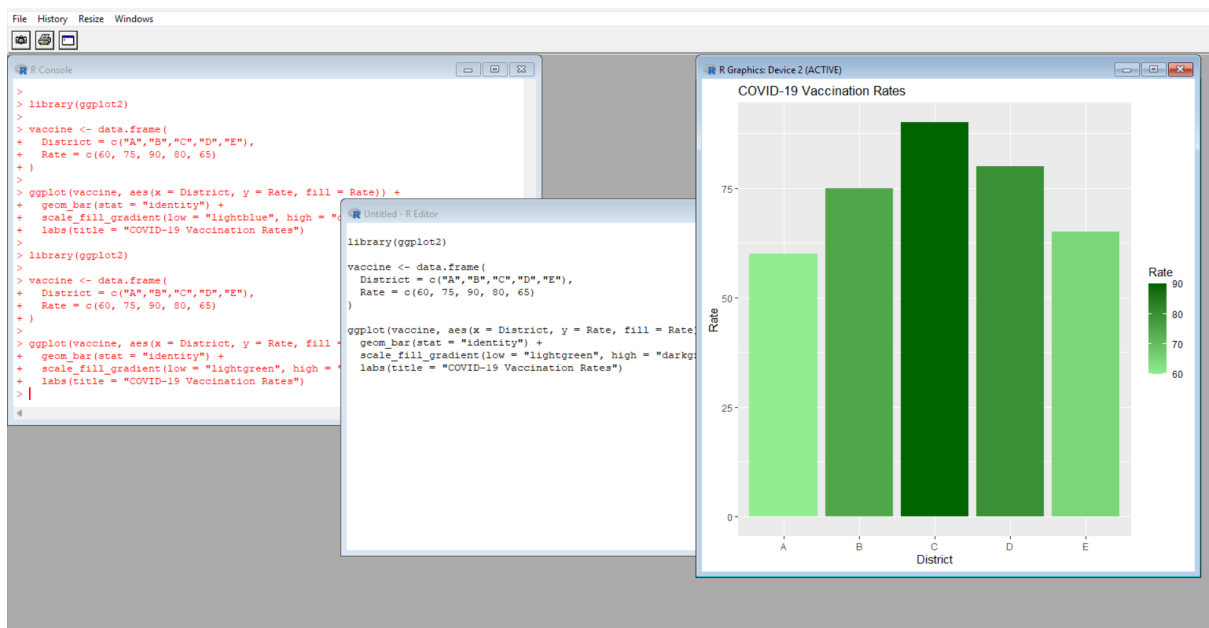


Version 2: Public Dashboard (Green Palette)

```
library(ggplot2)
```

```
vaccine <- data.frame(  
  District = c("A","B","C","D","E"),  
  Rate = c(60, 75, 90, 80, 65)  
)
```

```
ggplot(vaccine, aes(x = District, y = Rate, fill = Rate)) +  
  geom_bar(stat = "identity") +  
  scale_fill_gradient(low = "lightgreen", high = "darkgreen") +  
  labs(title = "COVID-19 Vaccination Rates")
```



Python codes

1.Smart City Air Quality Monitoring

```
import matplotlib.pyplot as plt
```

```
# Sample data
```

```
location = [1, 2, 3, 4, 5]
```

```
pm25 = [30, 60, 90, 50, 80]
```

```
level = ["Low", "Moderate", "High", "Moderate", "High"]
```

```
# 1. Sequential Palette
```

```
plt.scatter(location, pm25, c=pm25, cmap="Blues", s=100)
```

```
plt.title("Sequential Palette")
```

```
plt.xlabel("Location")
```

```
plt.ylabel("PM2.5")
```

```
plt.colorbar(label="PM2.5")
```

```
plt.show()
```

```
# 2. Diverging Palette
```

```
plt.scatter(location, pm25, c=pm25, cmap="coolwarm", s=100)
```

```
plt.title("Diverging Palette")
```

```
plt.xlabel("Location")
```

```
plt.ylabel("PM2.5")
```

```
plt.colorbar(label="PM2.5")
```

```
plt.show()
```

```
# 3. Qualitative Palette
```

```
colors = ["green", "orange", "red", "orange", "red"]
```

```
plt.scatter(location, pm25, c=colors, s=100)
```

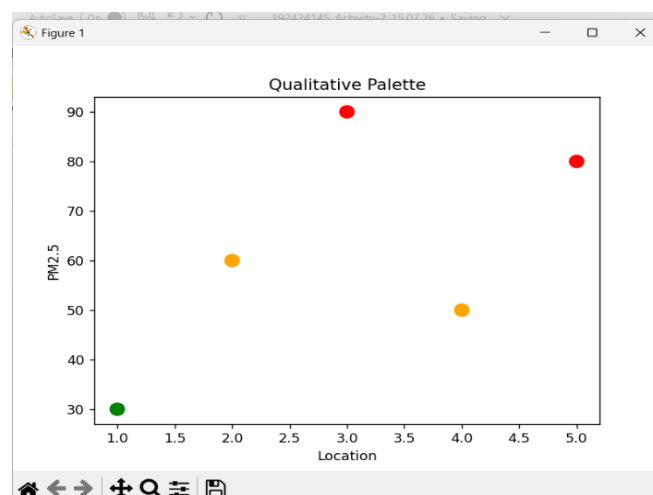
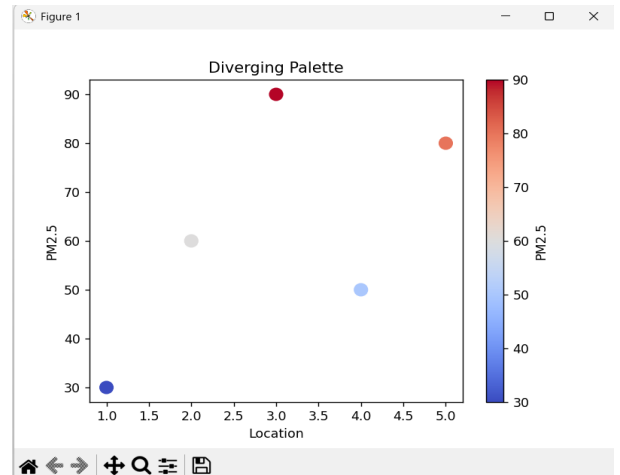
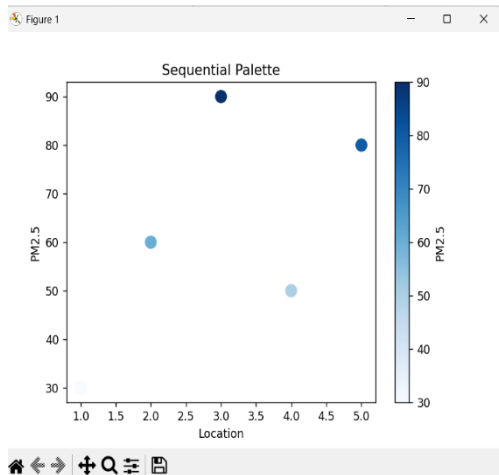
```
plt.title("Qualitative Palette")
```



```
plt.xlabel("Location")
```

```
plt.ylabel("PM2.5")
```

```
plt.show()
```



2. Earthquake Risk Analysis

```
import matplotlib.pyplot as plt
```

```
# Sample data
```

```
magnitude = [0.5, 1, 2, 3, 5, 7, 9]
```

```
depth = [20, 40, 80, 150, 250, 500, 700]
```

```
# Linear Scale
```

```
plt.scatter(magnitude, depth, color="blue")
```

```
plt.title("Earthquake Risk Analysis (Linear Scale)")
```

```
plt.xlabel("Magnitude")
```

```
plt.ylabel("Depth (km)")
```

```
plt.show()
```

```
# Logarithmic Scale
```

```
plt.scatter(magnitude, depth, color="red")
```

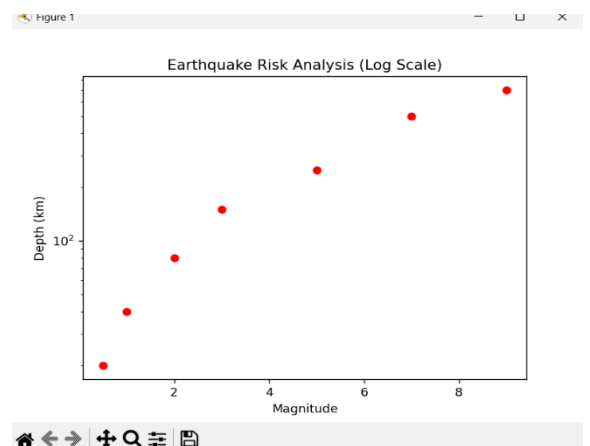
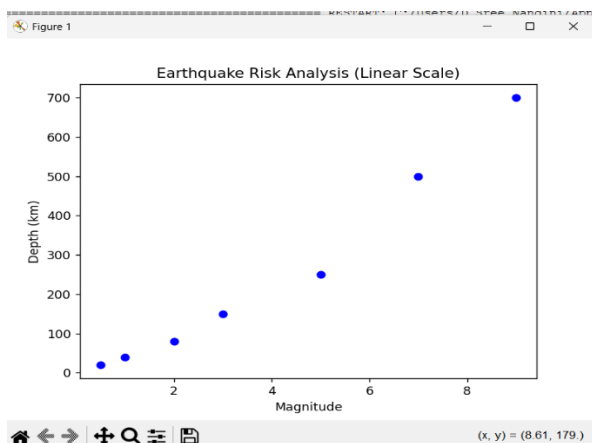
```
plt.yscale("log")
```

```
plt.title("Earthquake Risk Analysis (Log Scale)")
```

```
plt.xlabel("Magnitude")
```

```
plt.ylabel("Depth (km)")
```

```
plt.show()
```



3. Wind Direction Analysis for Airport Operations

Cartesian Plot

```
import matplotlib.pyplot as plt
```

```
direction = ["N", "NE", "E", "SE", "S", "SW", "W", "NW"]
```

```
speed = [12, 18, 10, 8, 15, 20, 14, 11]
```

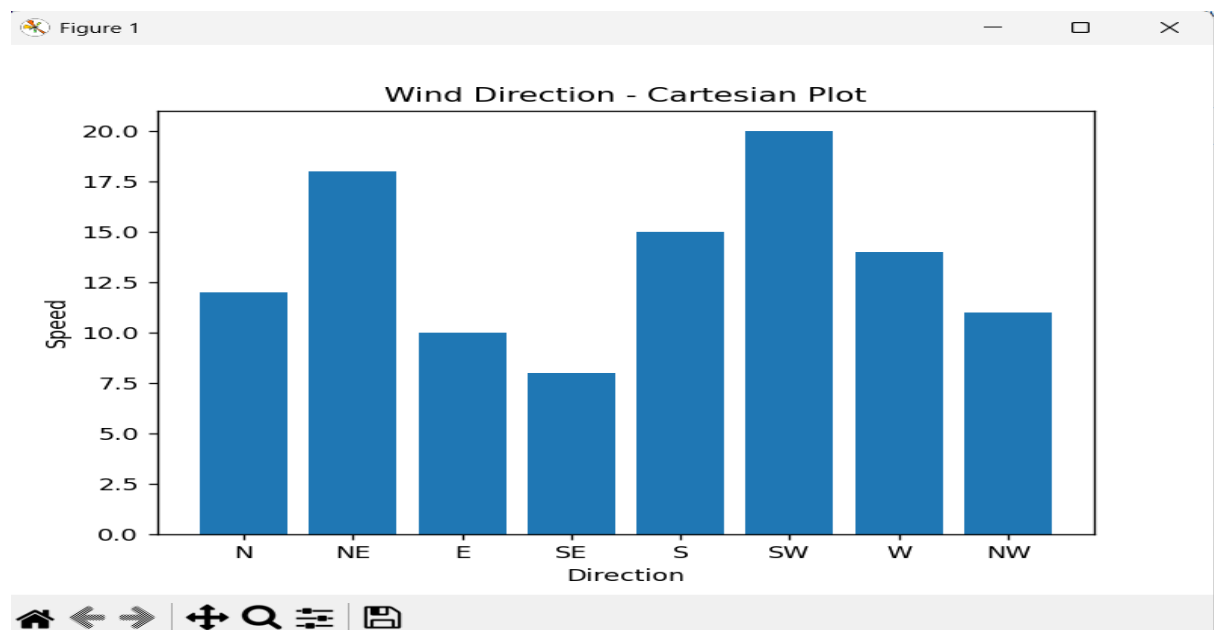
```
plt.bar(direction, speed)
```

```
plt.title("Wind Direction - Cartesian Plot")
```

```
plt.xlabel("Direction")
```

```
plt.ylabel("Speed")
```

```
plt.show()
```



Polar Plot (Wind Rose)

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
speed = [12, 18, 10, 8, 15, 20, 14, 11]
```

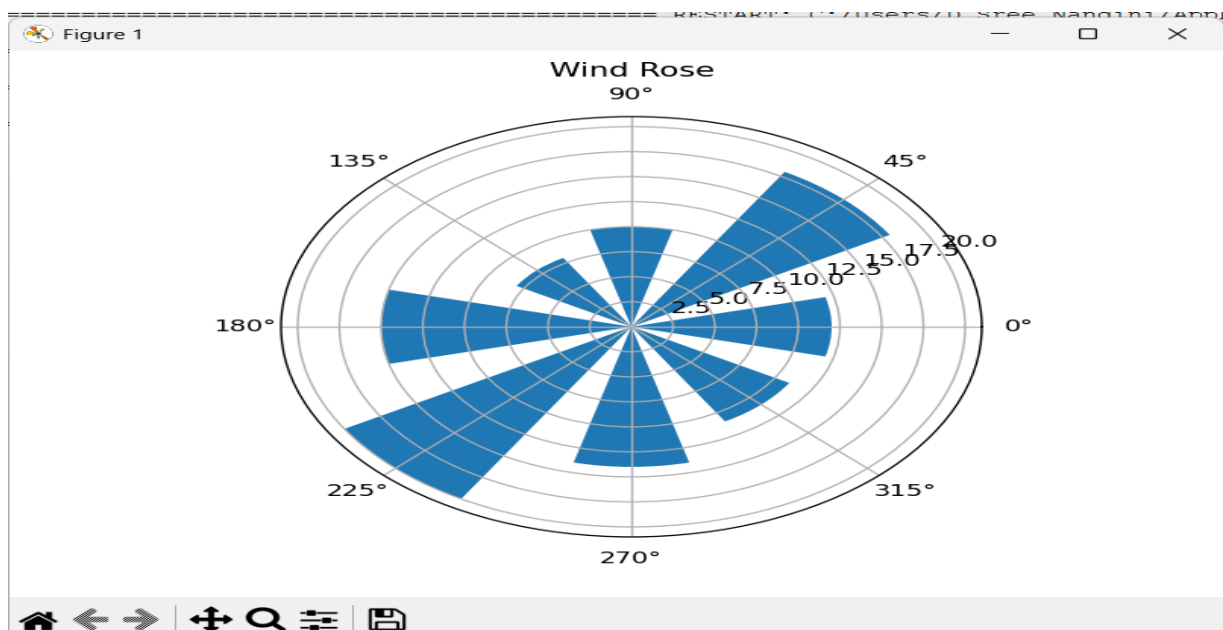
```
theta = np.linspace(0, 2*np.pi, len(speed), endpoint=False)
```

```
ax = plt.subplot(111, polar=True)
```

```
ax.bar(theta, speed, width=0.5)
```

```
plt.title("Wind Rose")
```

```
plt.show()
```



4. Hospital Patient Monitoring Dashboard

```
import matplotlib.pyplot as plt
```

```
parameters = ["Heart Rate", "Oxygen", "BP", "Respiration", "Temperature"]
```

```
values = [130, 88, 140, 28, 39]
```

```
colors = ["red", "green", "red", "orange", "red"]
```

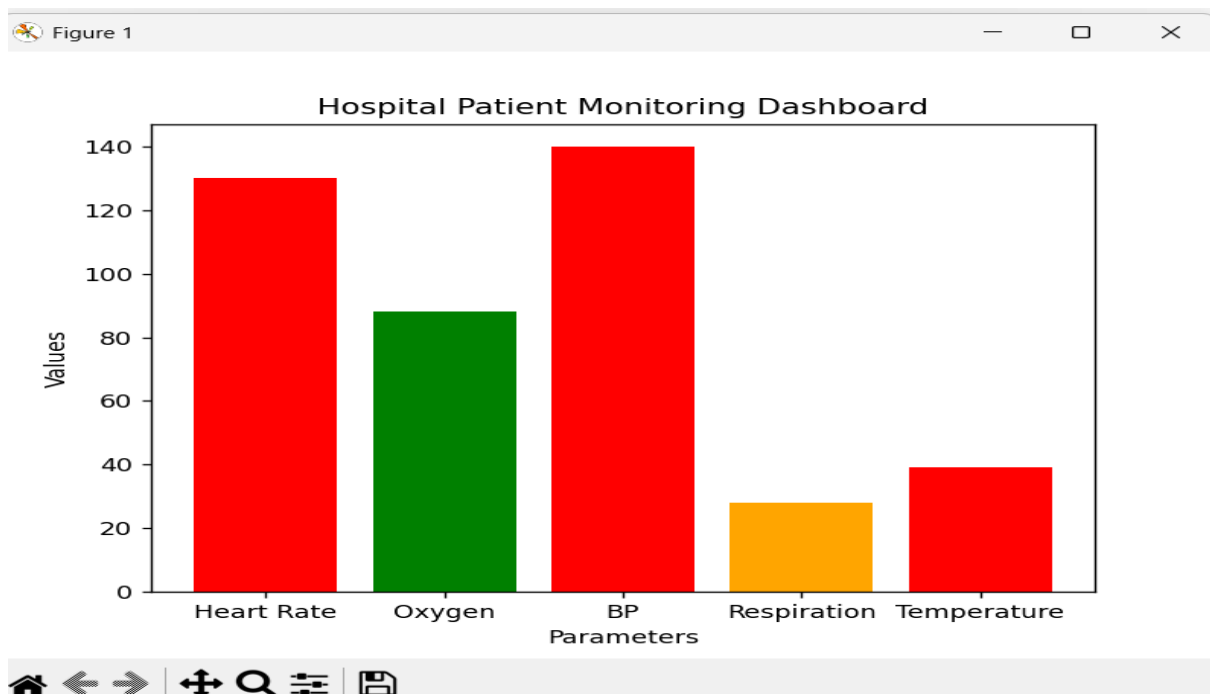
```
plt.bar(parameters, values, color=colors)
```

```
plt.title("Hospital Patient Monitoring Dashboard")
```

```
plt.xlabel("Parameters")
```

```
plt.ylabel("Values")
```

```
plt.show()
```



5. Global Climate Change Study

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
data = np.array([  
    [-1.5, -0.5, 0.2],  
    [0.5, 1.0, 1.8],  
    [-0.8, 0.3, 2.0]  
])
```

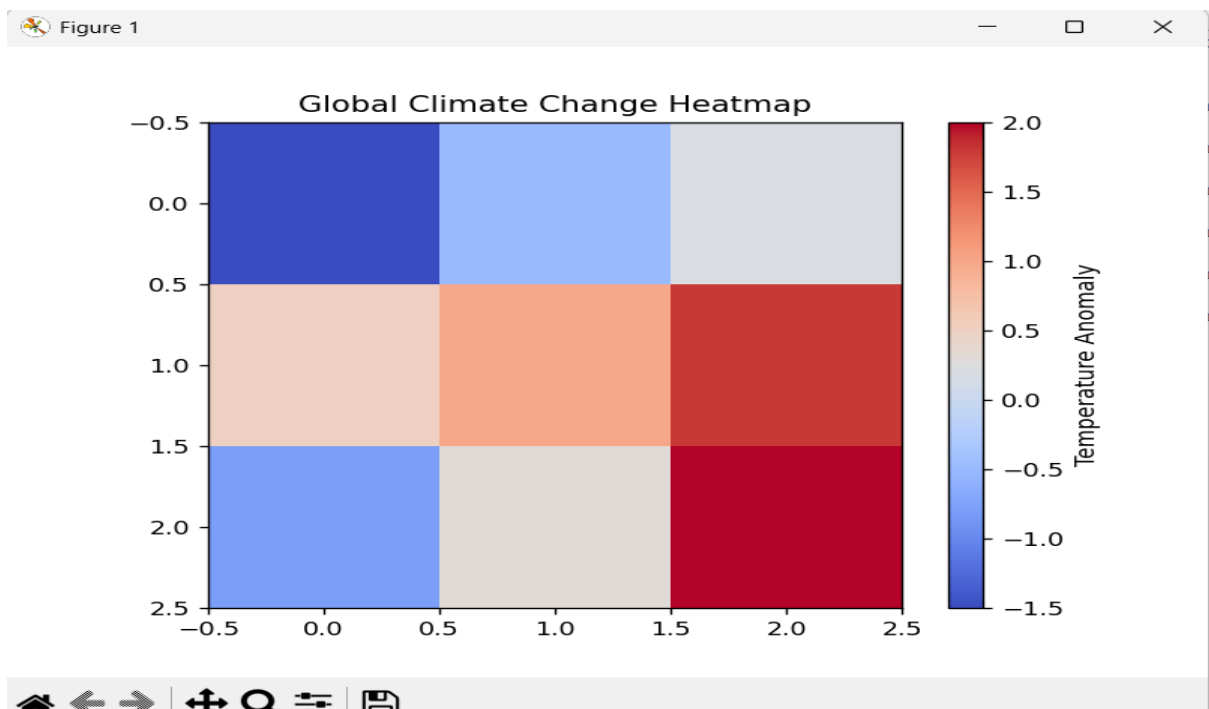
```
# Heatmap
```

```
plt.imshow(data, cmap="coolwarm")
```

```
plt.colorbar(label="Temperature Anomaly")
```

```
plt.title("Global Climate Change Heatmap")
```

```
plt.show()
```



6. E-Commerce Sales Analytics

```
import matplotlib.pyplot as plt
```

```
category = ["Electronics", "Clothing", "Furniture", "Toys"]
```

```
sales = [500, 300, 400, 600]
```

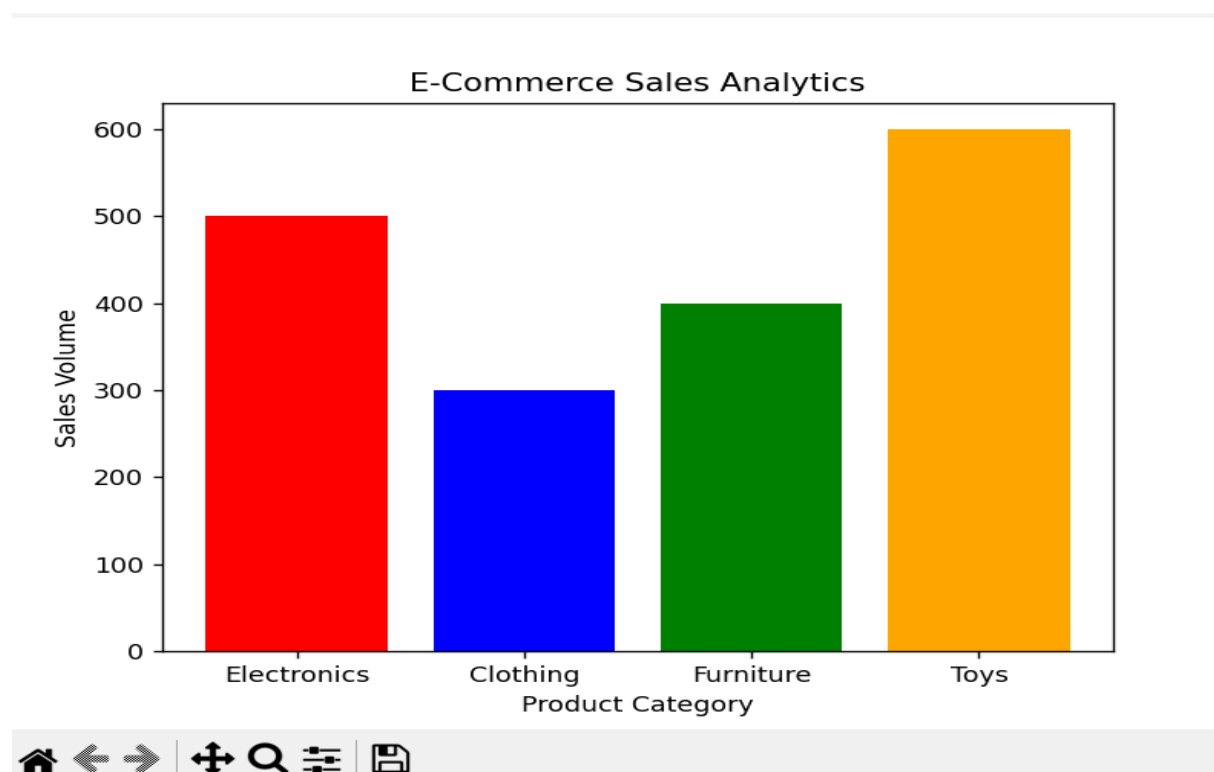
```
plt.bar(category, sales, color=["red", "blue", "green", "orange"])
```

```
plt.title("E-Commerce Sales Analytics")
```

```
plt.xlabel("Product Category")
```

```
plt.ylabel("Sales Volume")
```

```
plt.show()
```



7. Satellite Image Vegetation Analysis

```
import matplotlib.pyplot as plt
```

```
area = ["A", "B", "C", "D", "E"]
```

```
ndvi = [-1, -0.5, 0, 0.5, 1]
```

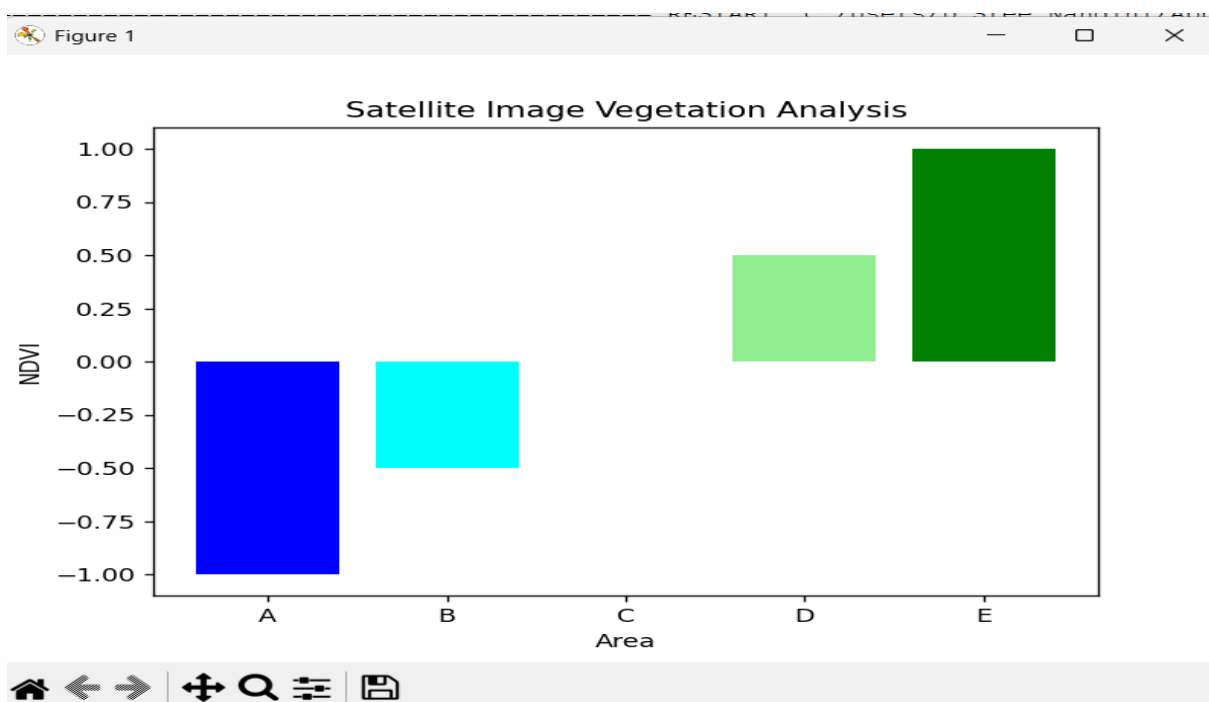
```
plt.bar(area, ndvi, color=["blue", "cyan", "yellow", "lightgreen", "green"])
```

```
plt.title("Satellite Image Vegetation Analysis")
```

```
plt.xlabel("Area")
```

```
plt.ylabel("NDVI")
```

```
plt.show()
```



8. Cryptocurrency Market Volatility

```
import matplotlib.pyplot as plt
```

```
day = [1, 2, 3, 4, 5]
```

```
bitcoin = [30000, 35000, 40000, 45000, 50000]
```

```
plt.plot(day, bitcoin, color="blue")
```

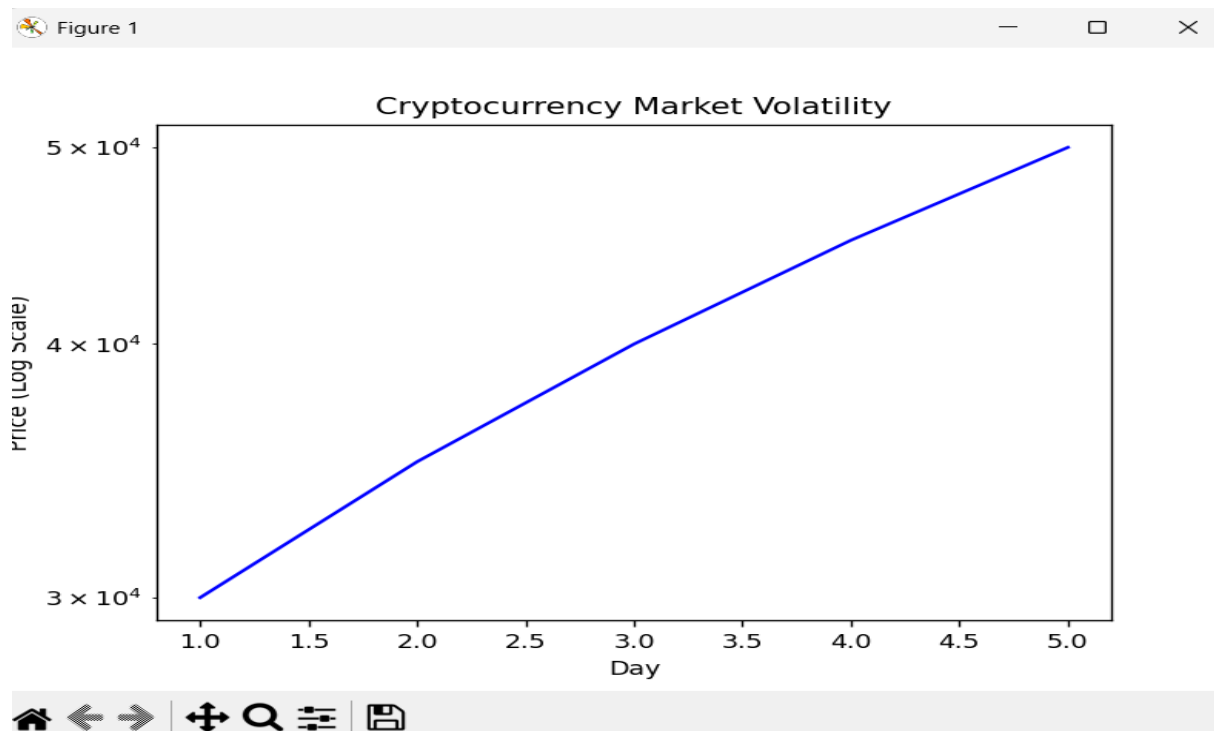
```
plt.yscale("log")
```

```
plt.title("Cryptocurrency Market Volatility")
```

```
plt.xlabel("Day")
```

```
plt.ylabel("Price (Log Scale)")
```

```
plt.show()
```



9. Marathon Runner Performance Analysis

Cartesian Plot

```
import matplotlib.pyplot as plt
```

```
distance = [5, 10, 15, 20, 25]
```

```
speed = [10, 12, 11, 13, 12]
```

```
plt.scatter(distance, speed, c=speed, cmap="viridis")
```

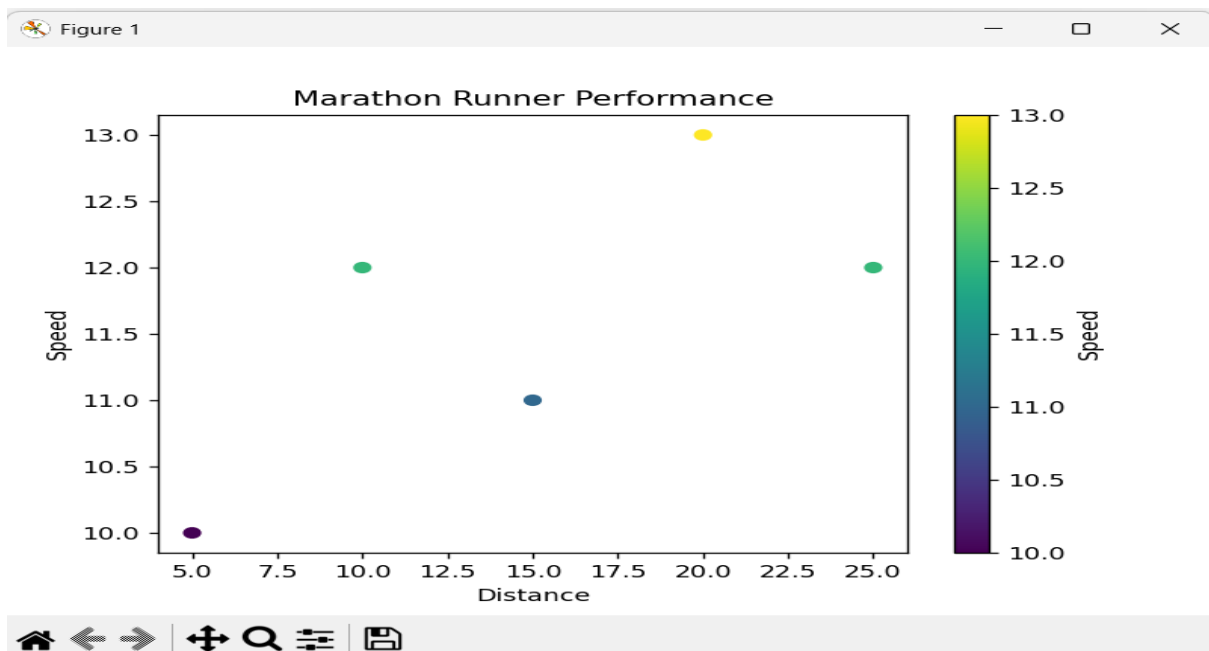
```
plt.colorbar(label="Speed")
```

```
plt.title("Marathon Runner Performance")
```

```
plt.xlabel("Distance")
```

```
plt.ylabel("Speed")
```

```
plt.show()
```



Polar Plot

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

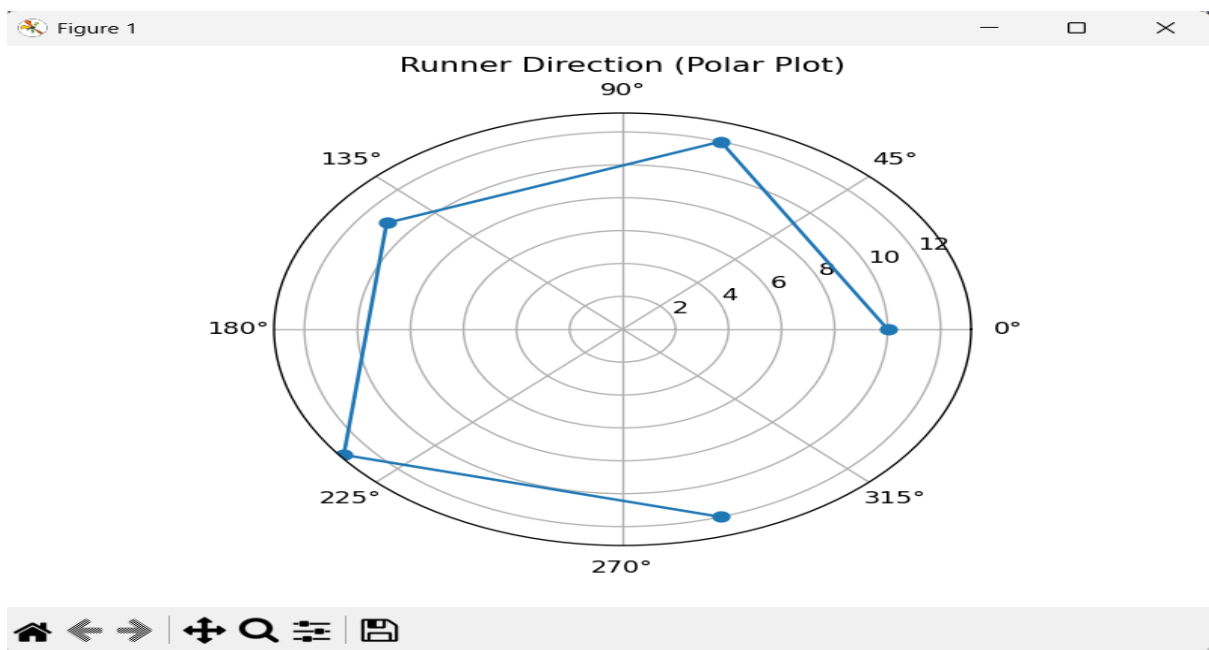
```
speed = [10, 12, 11, 13, 12]
```

```
angle = np.linspace(0, 2*np.pi, len(speed), endpoint=False)
```

```
plt.polar(angle, speed, 'o-')
```

```
plt.title("Runner Direction (Polar Plot)")
```

```
plt.show()
```



10. COVID-19 Vaccination Campaign Dashboard

Version 1 – Blue Color Palette (For Policymakers)

```
import matplotlib.pyplot as plt
```

```
district = ["A", "B", "C", "D", "E"]
```

```
rate = [60, 75, 90, 80, 65]
```

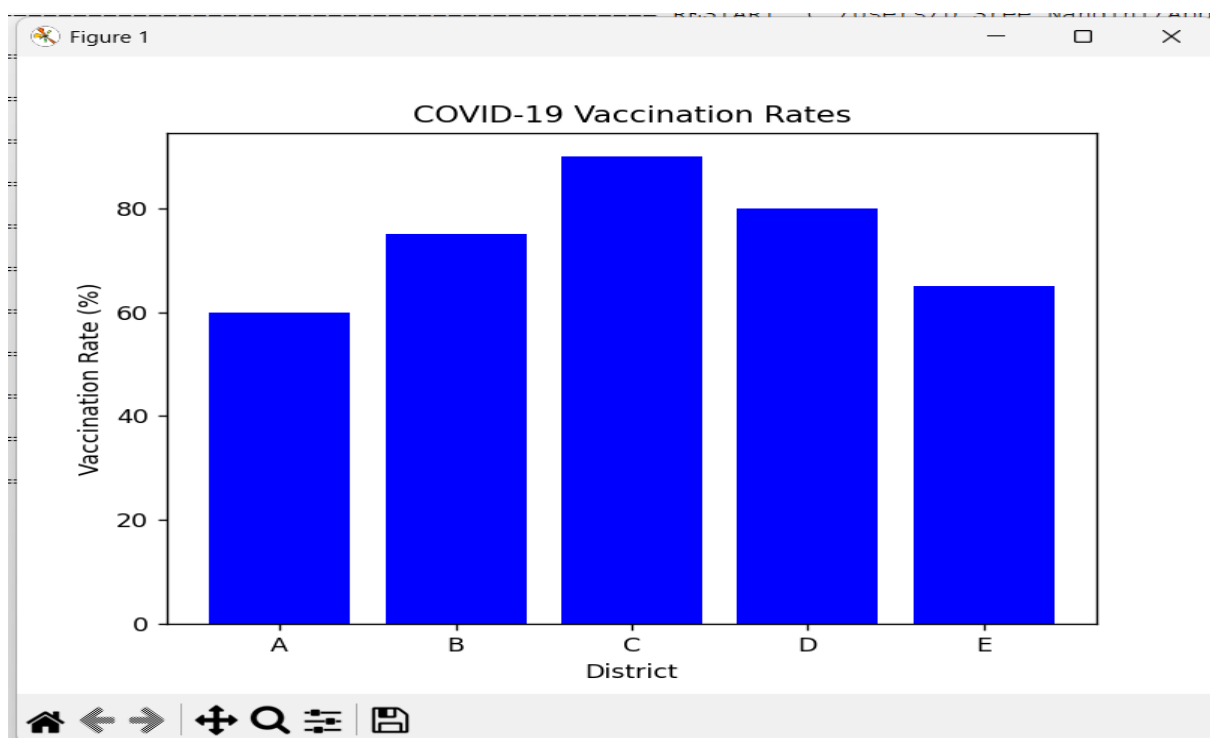
```
plt.bar(district, rate, color="blue")
```

```
plt.title("COVID-19 Vaccination Rates")
```

```
plt.xlabel("District")
```

```
plt.ylabel("Vaccination Rate (%)")
```

```
plt.show()
```



Version 2 – Green Color Palette (For General Public)

```
import matplotlib.pyplot as plt
```

```
district = ["A", "B", "C", "D", "E"]
```

```
rate = [60, 75, 90, 80, 65]
```

```
plt.bar(district, rate, color="green")
```

```
plt.title("COVID-19 Vaccination Rates")
```

```
plt.xlabel("District")
```

```
plt.ylabel("Vaccination Rate (%)")
```

```
plt.show()
```

